

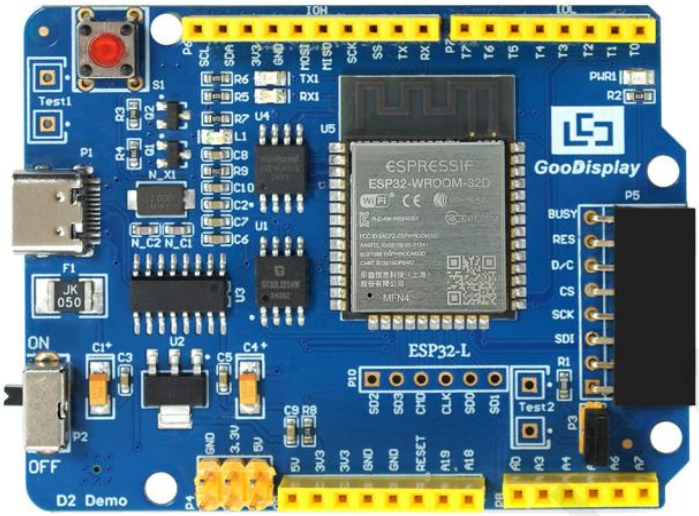
# SPI Full Series E-Paper WiFi Local and Remote Driver User Manual

GooDisplay®

## ESP32-L

Dalian Good Display Co., Ltd.

# Product Specifications




| | |
|---|---|
| Customer | Standard |
| Description | ePaper Display Driver Board |
| Model Name | ESP32-L |
| Date | 2025/02/12 |
| Revision | 1.0 |

| | Design Engineering | | |
|---|---|---|---|
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# 1. Product Overview

ESP32-L e-paper display driver board uses the popular WiFi communication method. It is also equipped with PC software and a mobile app, allowing users to update the current display content directly from a PC or a mobile device. This design simplifies the update process, as users do not need to perform image encoding operations and only need to import the images to be displayed into the software. Through the PC software and mobile app, users can achieve real-time updates of the content on the e-paper display driver board, which is especially useful for application scenarios that require frequent changes to display content or real-time information updates. By adopting WiFi communication, the operation becomes more convenient and does not require professional skills, enabling ordinary users to easily update and control the displayed content.

# 2. Product Advantages

- The display content can be updated via WiFi
- No image encoding is required, truly enabling one-click updates
- Supports image file formats: *.JPG, *.BMP, *.PNG

# 3. Product Specifications

| | |
|---|---|
| Model | ESP32-L |
| Display Parameters | Screen size: 0.97 inch to 13.3 inch<br>Display type: black and white / black, white and red / black, white, red and yellow e-paper display<br>Resolution: 184 × 88 to 960 × 680 |
| Interface Parameters | USB interface: supports USB network configuration<br>Supports firmware update via USB |

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| | |
|---|---|
| Power Consumption Parameters | Operating voltage: 5V<br>Refresh power consumption: <500 mW |
| Software Parameters | Image refresh time:<br>① Black and white e-paper display: 3 s<br>② Black, white and red e-paper display: 18 s<br>③ Black, white, red and yellow e-paper display: 22 s<br>Image update method: WiFi update<br>Language: English by default |
| Main Board Parameters | WiFi frequency band: 2.4 GHz (5G not supported)<br>MCU: ESP32-WROOM-32D<br>RAM: 520 KB<br>ROM: 448 KB |
| Physical Specifications | Weight: approximately 204 g<br>Dimensions: 68 × 63 × 1.6 mm |
| Temperature and Humidity Parameters | Operating temperature: 0°C to 50°C<br>Storage temperature: -25°C to 70°C<br>Operating humidity: 40% to 70% |

## 4. Product Structure

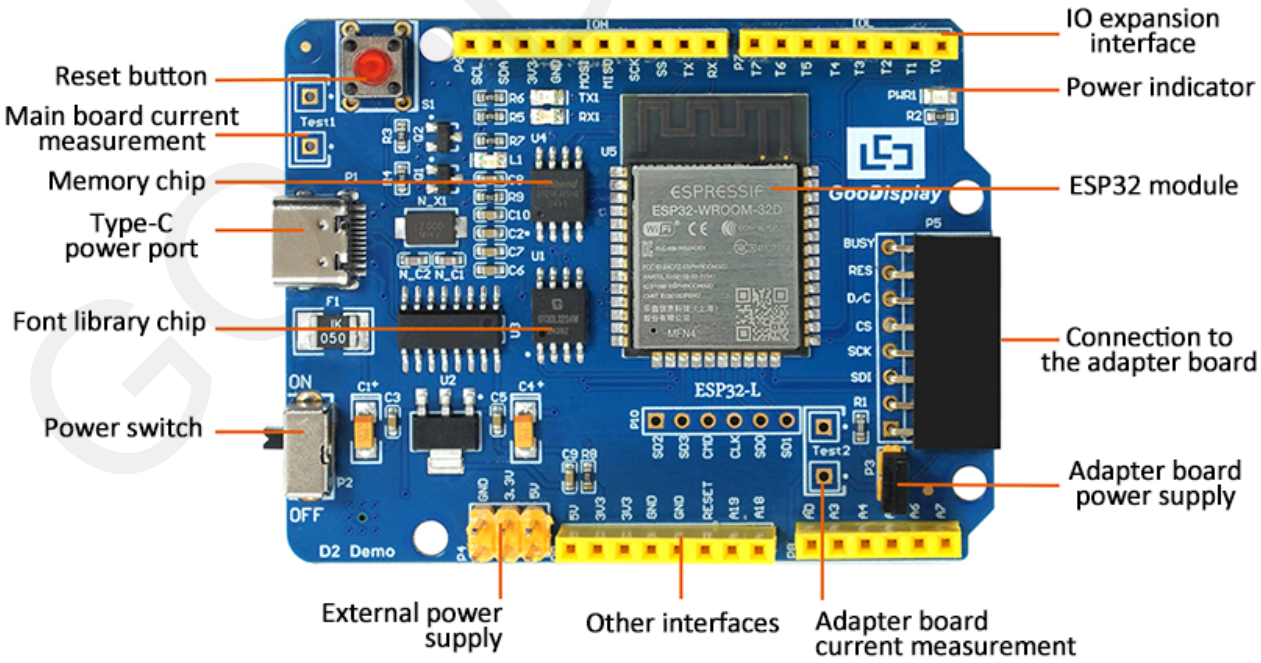



Figure 1. ESP32-L Front View



Figure 2. ESP32-L Rear View

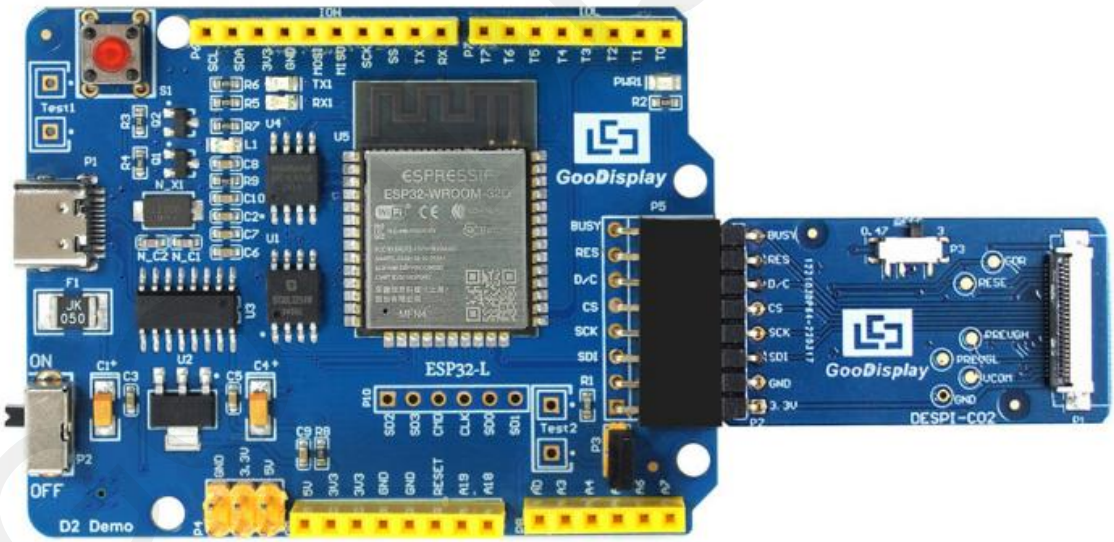



Figure 3. ESP32-L Connected to the Adapter Board

Download links for all the following materials:
https://github.com/gooddisplayshare/ESP32-L

# 5. LAN Display Update on PC

## 5.1. Image Requirements

1.File formats: BMP, JPG, PNG.

Resolution: select according to the corresponding screen size.

0.97: 184x88

1.54L: 152x152

1.54: 200x200

2.13: 250x122

2.66: 296x152

2.66H: 360x184

2.7: 264x176

2.9: 296x128

2.9H: 384x168

3.7: 416x240

4.2: 400x300

7.5: 800x480

13.3: 960x680

Use the LAN WiFi image transfer tool ImageToWiFi to import the image into the software. The following example uses a 7.5-inch monochrome display; operations for other sizes are similar.

## 5.2.Local Firmware Flashing

Download and flash the local firmware EpdAllWiFi. For the firmware flashing method, please refer to the ESP32-L User Manual.

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### 5.3.WiFi Network Settings

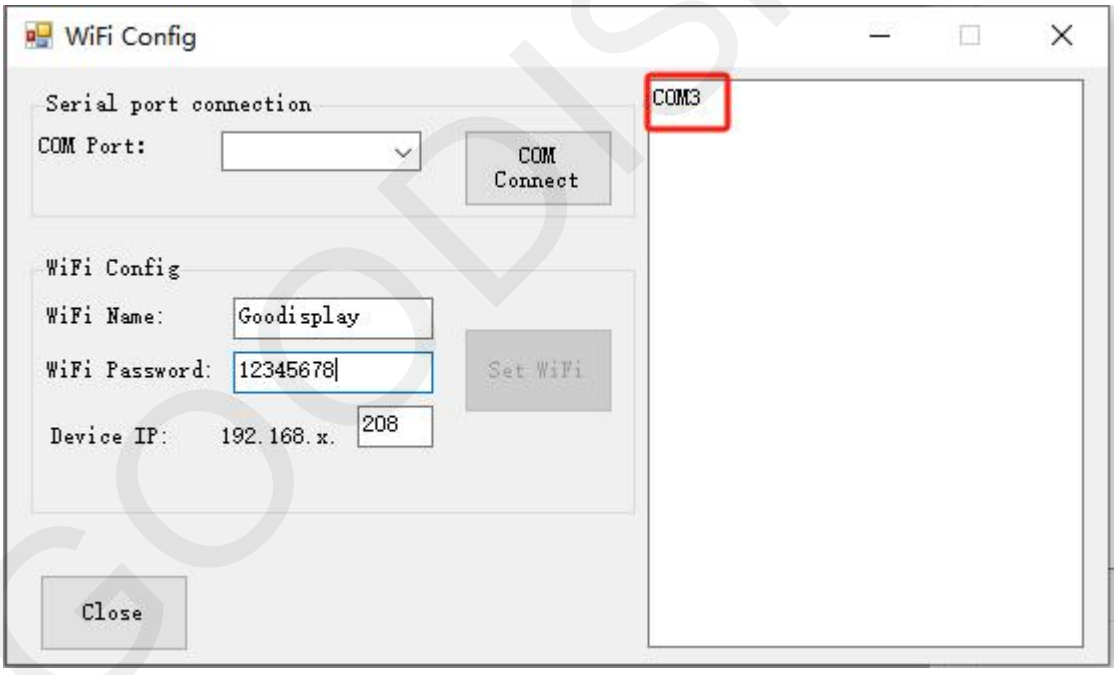
1) Device firmware WiFi settings: Users need to configure the WiFi SSID, password, and IP number through the ImageToWiFi software. The default IP number is 208. The default WiFi name is Goodisplay, and the WiFi password is 12345678. In addition, without modifying the WiFi settings, users can directly create a WiFi hotspot on a mobile phone with the same name and password as the device's default parameters, and then use the mobile phone to update the device content directly.

Note: Please select 2.4 GHz WiFi. 5 GHz WiFi is not supported at present.

2) The CH340 driver must be installed on the PC first. Installing the driver is required; otherwise, the device USB interface cannot be used.

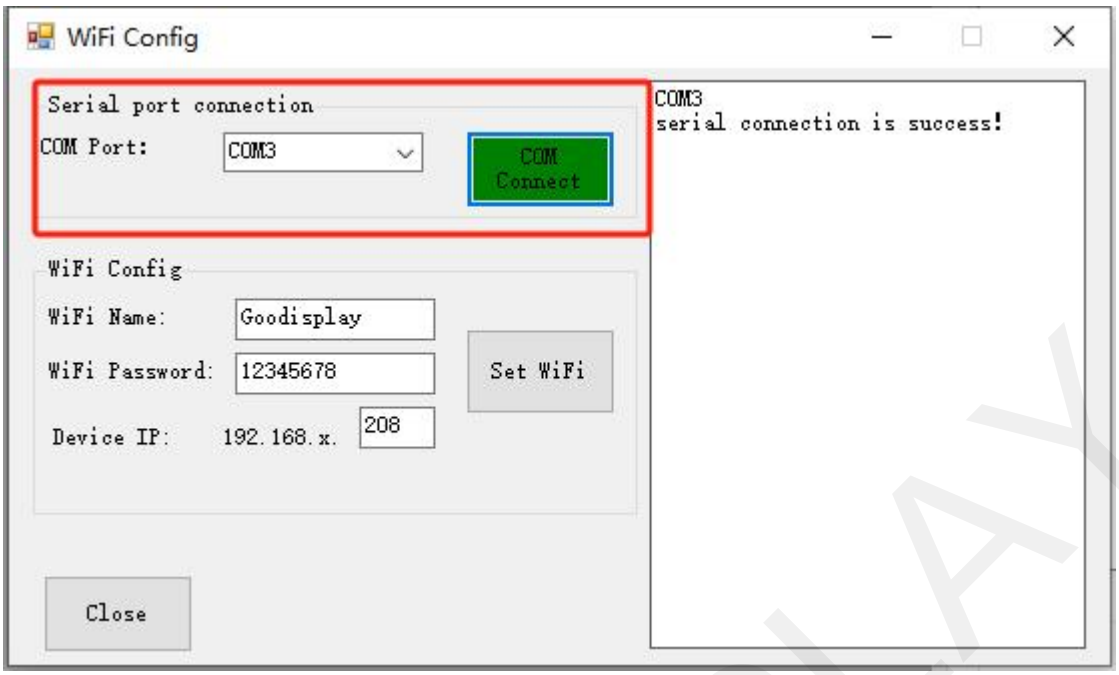
Open the ImageToWiFi software and locate the 

button to enter the WiFi serial port configuration interface. The status bar on the right will display the USB port number of the currently detected device.



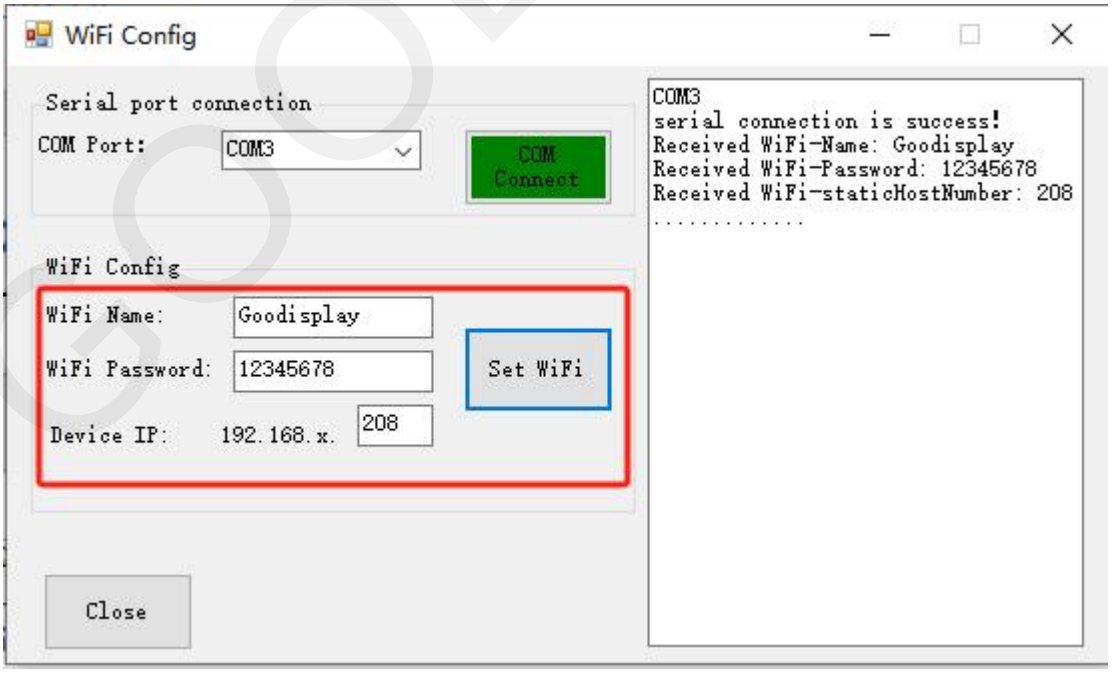

3) Select the device USB port number and click the 

button. When the USB port is successfully connected, the button will turn green, and the status bar will display that the serial port connection is successful.

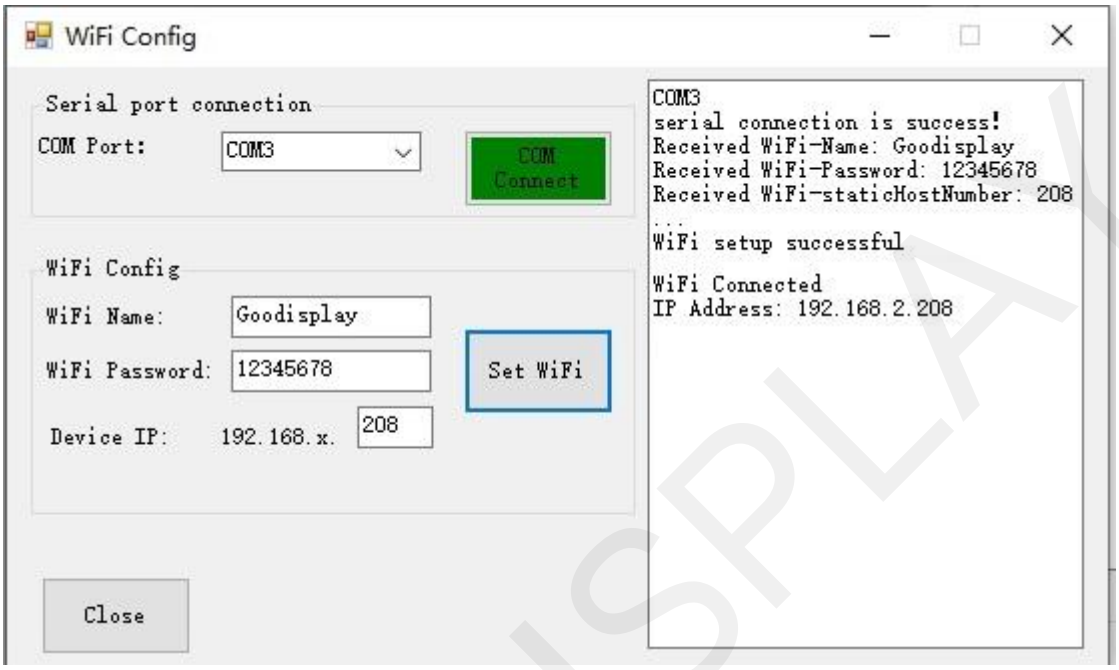



4) In the WiFi Config section, set the current WiFi SSID, password, and the IP number of this device. It is recommended to choose an IP number above 200 to avoid conflicts with other network devices. After completing the above information, click the 

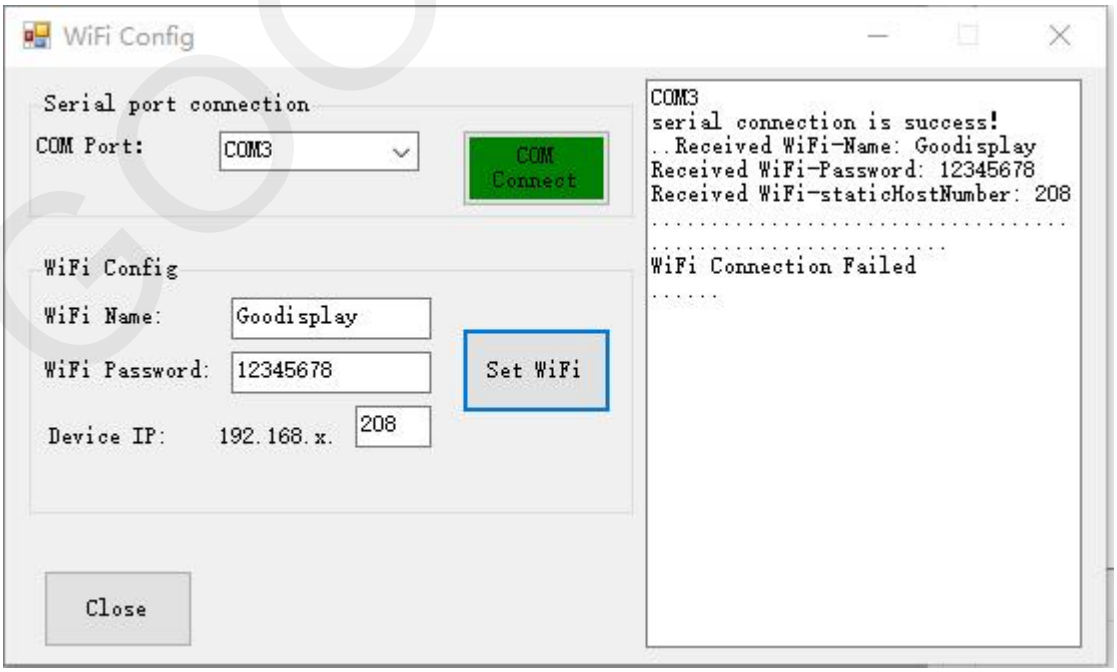
button to send the WiFi configuration information to the device. The status bar will simultaneously display the device network configuration status.



5） If the WiFi connection is successful, the status bar will display the device's WiFi connection status and show the current IP address of the device. At the same time, the device will also display the current WiFi connection status.




6） If the status bar shows that the WiFi connection failed, please check whether the WiFi configuration information is correct. The device will also display the current WiFi connection status.



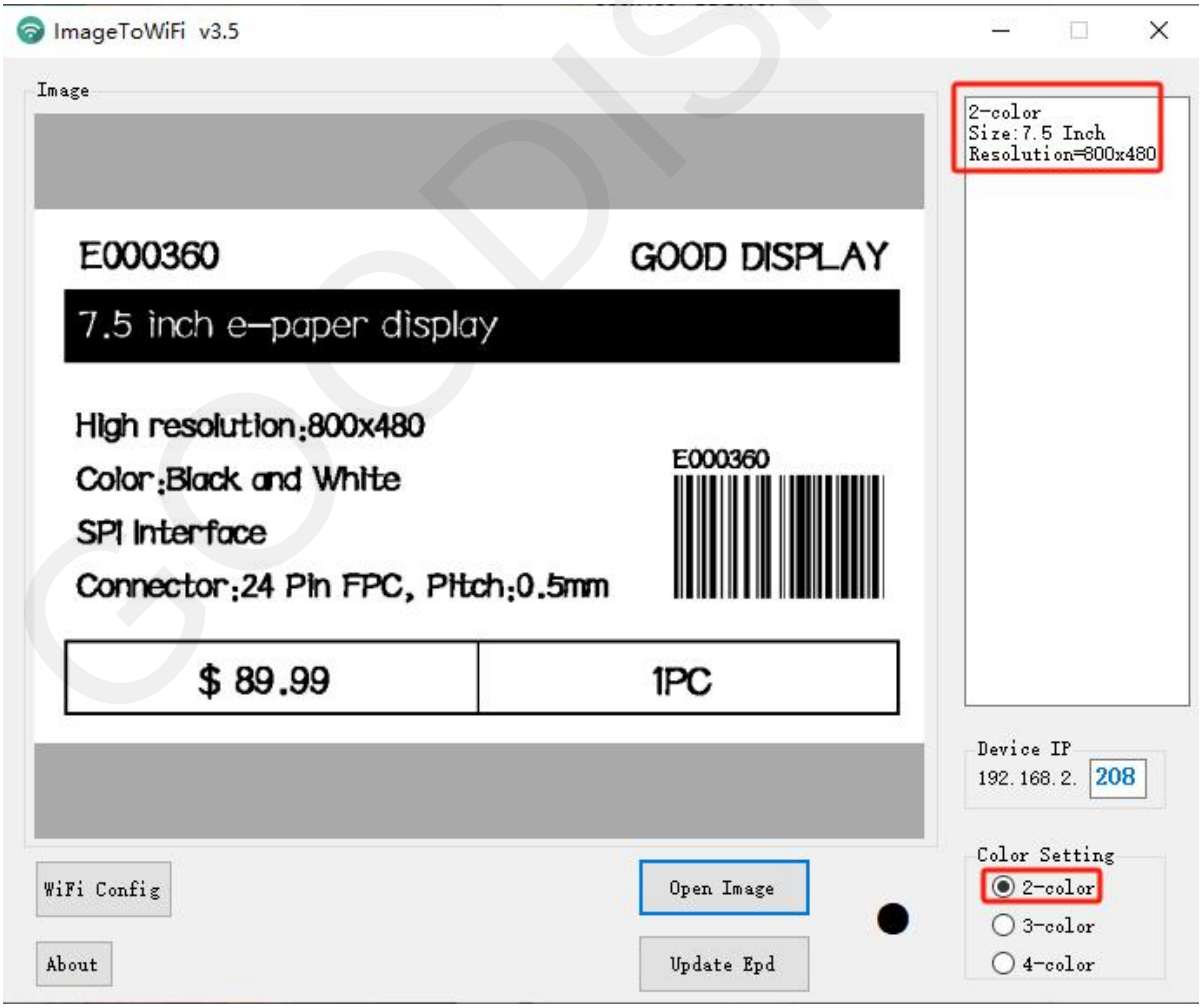
### 5.4. Image Display

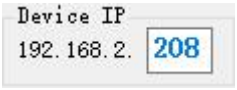
The image display section takes the black and white e-paper display as an example. For other color e-paper displays, select the corresponding color mode.

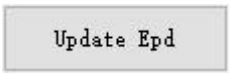
Among them, 2-color refers to the black and white e-paper display, 3-color refers to the black, white and red e-paper display, and 4-color refers to the black, white, red and yellow e-paper display.

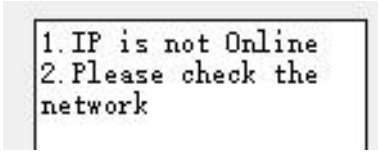
1. Power on the device. It is recommended to use a power supply of 5V 1A or above.

2. Open the ImageToWiFi software, click the "Open File" button, and select the prepared image with a resolution of 800 × 480. After importing the image, the software will display the image size, resolution, and other parameters. Select "2-color".



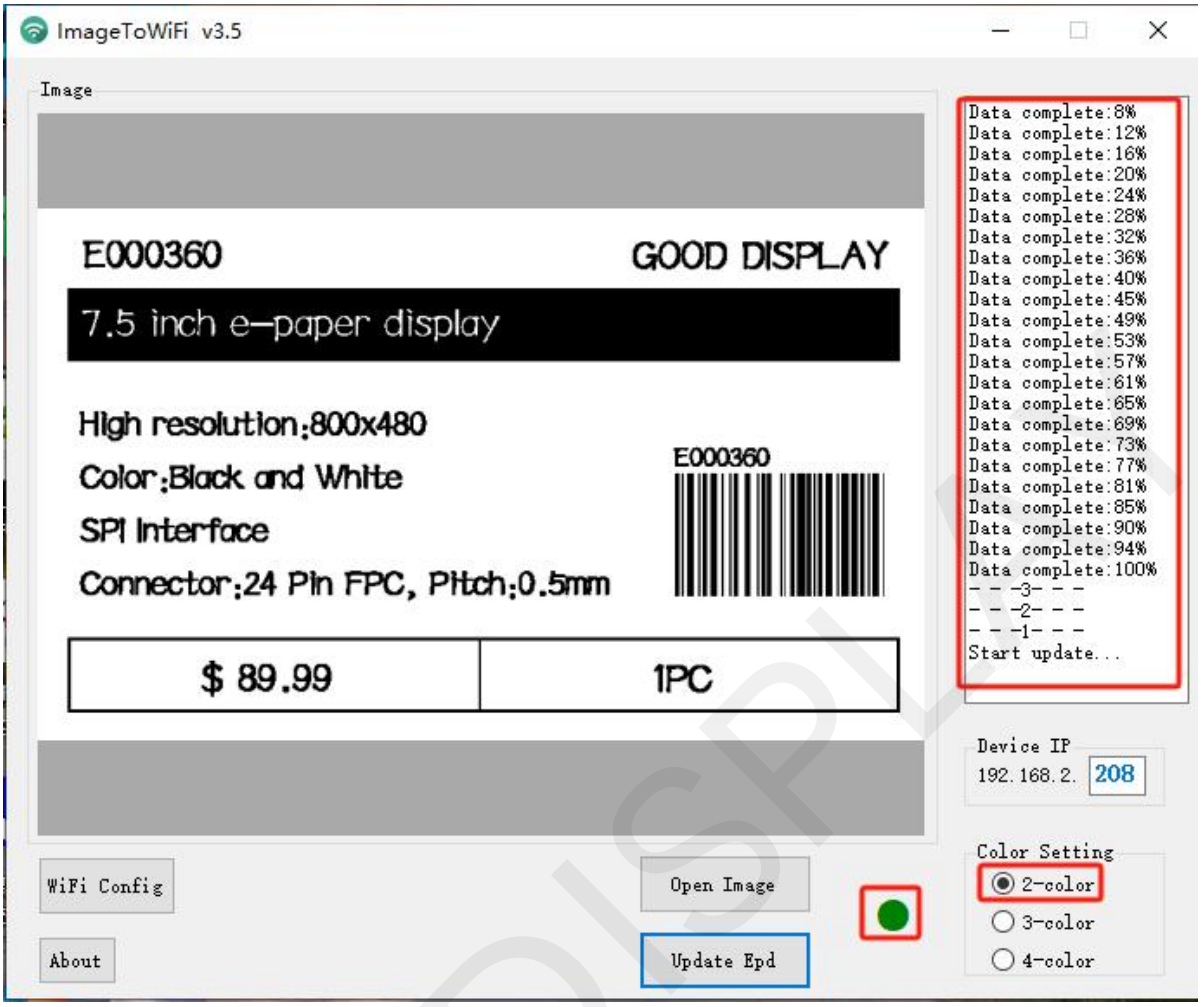
3. Set the Device IP in the software: The software Device IP must be consistent with the device IP number, which is 208 by default. 


4. Click the 

button below. The software will automatically connect to the online device. If the device is online, the indicator below will turn green  , and data transmission will begin. When the data transmission is normal, the information panel on the left will display the current data transmission progress. After the data transmission is completed, the software will display "Start update". 

If the device is not online, the status bar will display the relevant information.

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### 5.5. Frequently Asked Questions

| Issue | Solution |
|---|---|
| Cannot find local IP address | The local computer is not connected to the network |
| Cannot find remote device | Incorrect WiFi SSID or password entered on the device |
| | The software IP number does not match the device IP number (default is 208) |
| | Insufficient power supply to the device |
| | The device is offline, click the "Update Epd" button to rescan the online device IP status |

# 6. Mobile Display Update

The image display section takes the 7.5-inch black and white e-paper display as an example. For other color e-paper displays, select the corresponding color mode.

Among them, 2-color refers to the black and white e-paper display, 3-color refers to the black, white and red e-paper display, and 4-color refers to the black, white, red and yellow e-paper display.

## 6.1 Image Requirements

File formats: BMP, JPG, PNG, etc.

Resolution: select according to the corresponding screen size.

0.97": 184 × 88

1.54L": 152 × 152

1.54": 200 × 200

2.13": 250 × 122

2.66": 296 × 152

2.66H": 360 × 184

2.7": 264 × 176

2.9": 296 × 128

2.9H": 384 × 168

3.7": 416 × 240

4.2": 400 × 300

7.5": 800 × 480

13.3": 960 × 680

Install the WiFi-W2 mobile app on an Android phone and import the image into the software.

## 6.2 WiFi Network Settings

For the device WiFi network settings, please refer to Section 5.2 for USB network configuration on the PC side.

## 6.3 Image Display

Power on the driver board. It is recommended to use a power supply of

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5V 1A or above.

Install the WiFi-W2 mobile app on the phone.

Open the WiFi-W2 app. Select 2-color, then select the corresponding screen size. Here, select 7.5 to enter the WiFi image transfer main interface. The main interface generally displays a test image for users to test.




The WiFi-W2 app supports three image processing modes: Camera, Bitmap, and Note. Camera uses photos stored on the phone, Bitmap uses images optimized for the device's screen resolution, and Note is a notepad function. Before using any of these three modes, the IP must be set to the current device's IP; the software default is 208.



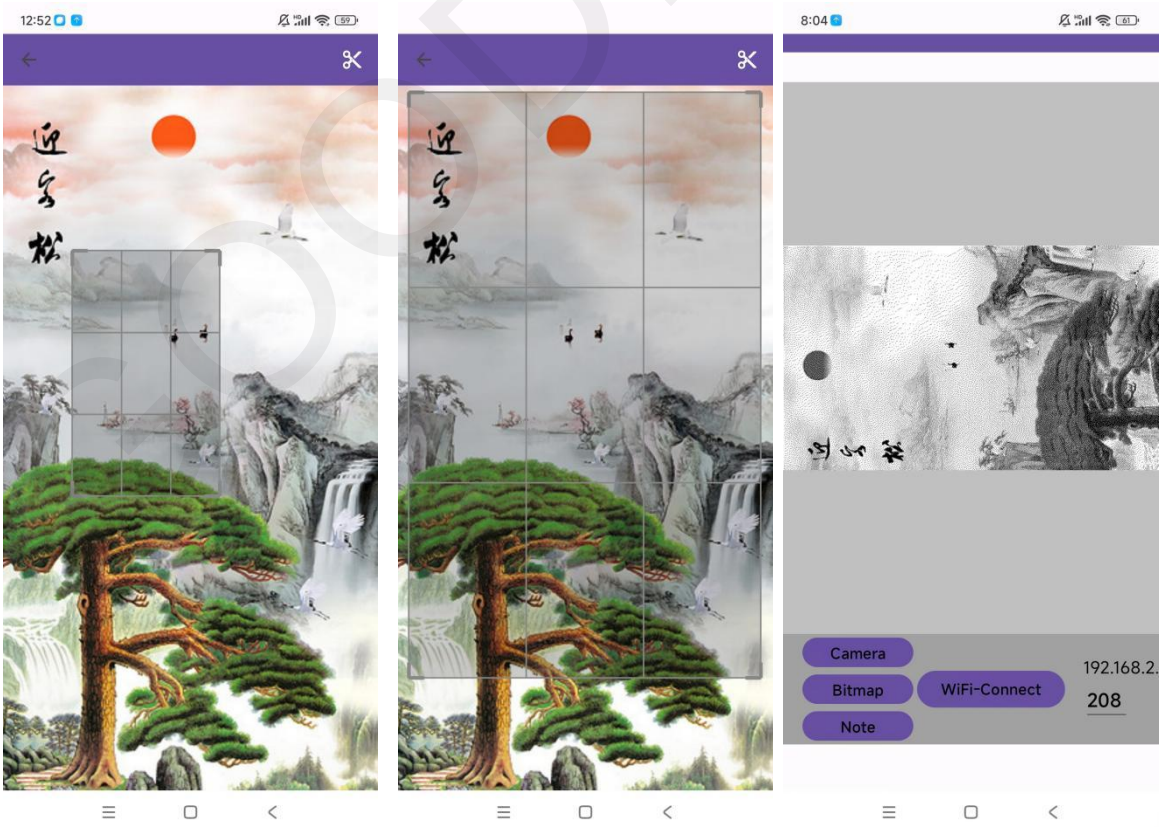
1) Camera Mode Image Transfer

Click the 

button to enter the phone's image selection interface. After selecting an image, the software will enter the image preprocessing interface, where you can crop the image. Drag the bottom-right corner of the gray rectangle to resize it, and place your finger in the center of the rectangle to move it freely. The area covered by the gray rectangle is the part that will be displayed. After selecting the rectangle, click the crop button  in the upper-right corner of the software. The image will be processed in the background and then enter the WiFi image transfer main interface. The larger the e-paper display, the longer the background processing will take, so please be patient.

Click the 

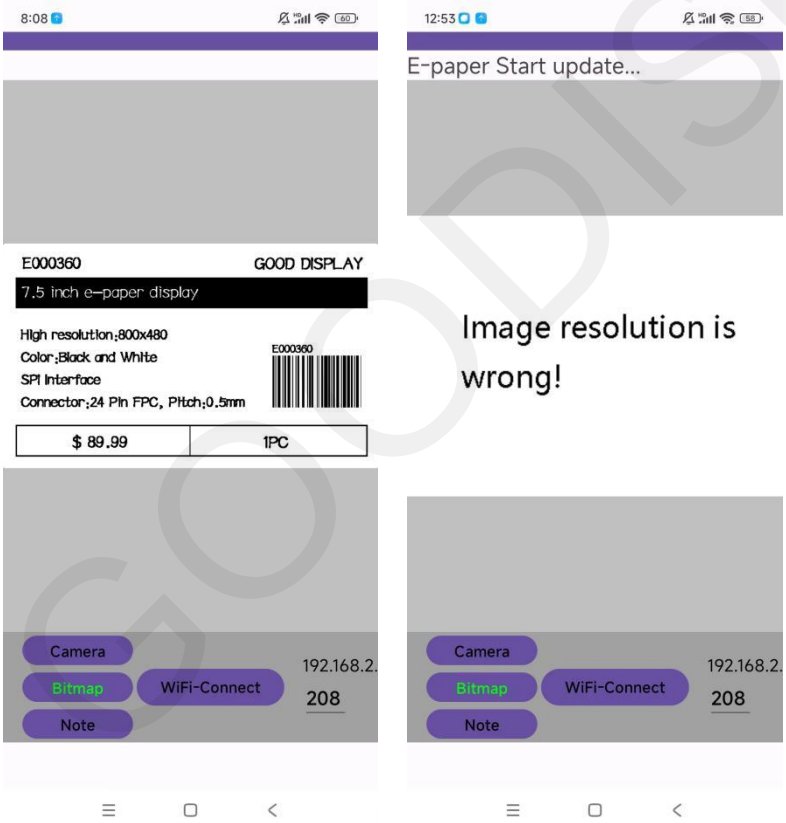
button on the WiFi image transfer main interface, and the image will be immediately sent to the WiFi device. The status bar at the top of the software will also display the data transmission progress and current working status.



2) Bitmap Mode Image Transfer

Click the Bitmap button to enter the specific image selection interface. Select an image with a resolution of 800 × 480, and the software will then enter the WiFi image transfer main interface. If the selected image resolution is incorrect, the WiFi image transfer main interface will display an error message.

Click the WiFi-Connect button on the WiFi image transfer main interface, and the image will be immediately sent to the WiFi device. The status bar at the top of the software will also display the data transmission progress and current working status.

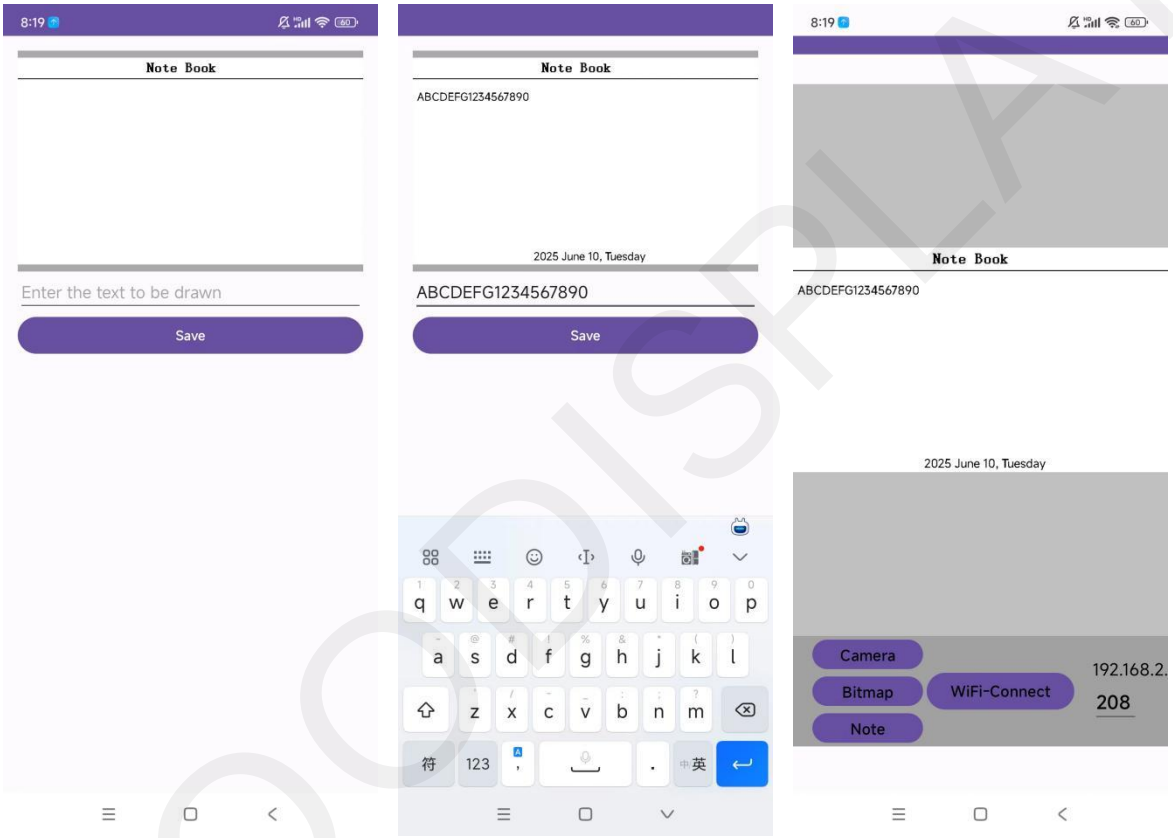



3) Note Mode Image Transfer

Click the Note button to enter the notepad editing interface. Users can manually enter the information to be recorded, then click

the Save button, and the software will enter the WiFi image transfer main interface.

Click the WiFi-Connect button on the WiFi image transfer main interface, and the image will be immediately sent to the WiFi device. The status bar at the top of the software will also display the data transmission progress and current working status.




## 7. Remote Update via Internet on PC

The image display section takes the 7.5-inch black and white e-paper display as an example. For other color e-paper displays, select the corresponding color mode. Among them, 2-color refers to the black and white e-paper display, 3-color refers to the black, white and red e-paper display, and 4-color refers to the black, white, red and yellow e-paper display.

### 7.1 Image Requirements

1) File formats: BMP, JPG, PNG, etc.

2) Resolution: select according to the corresponding screen size.

0.97": 184 × 88
1.54L": 152 × 152
1.54": 200 × 200
2.13": 250 × 122
2.66": 296 × 152
2.66H": 360 × 184
2.7": 264 × 176
2.9": 296 × 128
2.9H": 384 × 168
3.7": 416 × 240
4.2": 400 × 300
7.5": 800 × 480
13.3": 960 × 680

3) Use the remote image upload software ImageToServer to import the image into the software.

### 7.2 Remote Firmware Flashing

1) E-paper model setting: Configure the remote e-paper parameters by setting the following parameters in the firmware.

```
/*
1.inch
0.97--97  1.54L--153  1.54--154  2.13--213  2.66--266  2.66H--267  2.7--270  2.9--290
2.9H--291  3.7--370  4.2--420  7.5--750  13.3--1330
2.color  2--2color  3--3-color  4--4color
3.ic    2--SSD  3--UC
*/
int epdInch=290;//E-paper size
int epdColor=2;//E-paper color
int epdIC=2;//E-paper IC  2--SSD,3--UC
```

2) Download and flash the remote firmware EpdAllServer. For the firmware flashing method, please refer to the ESP32-L User Manual.

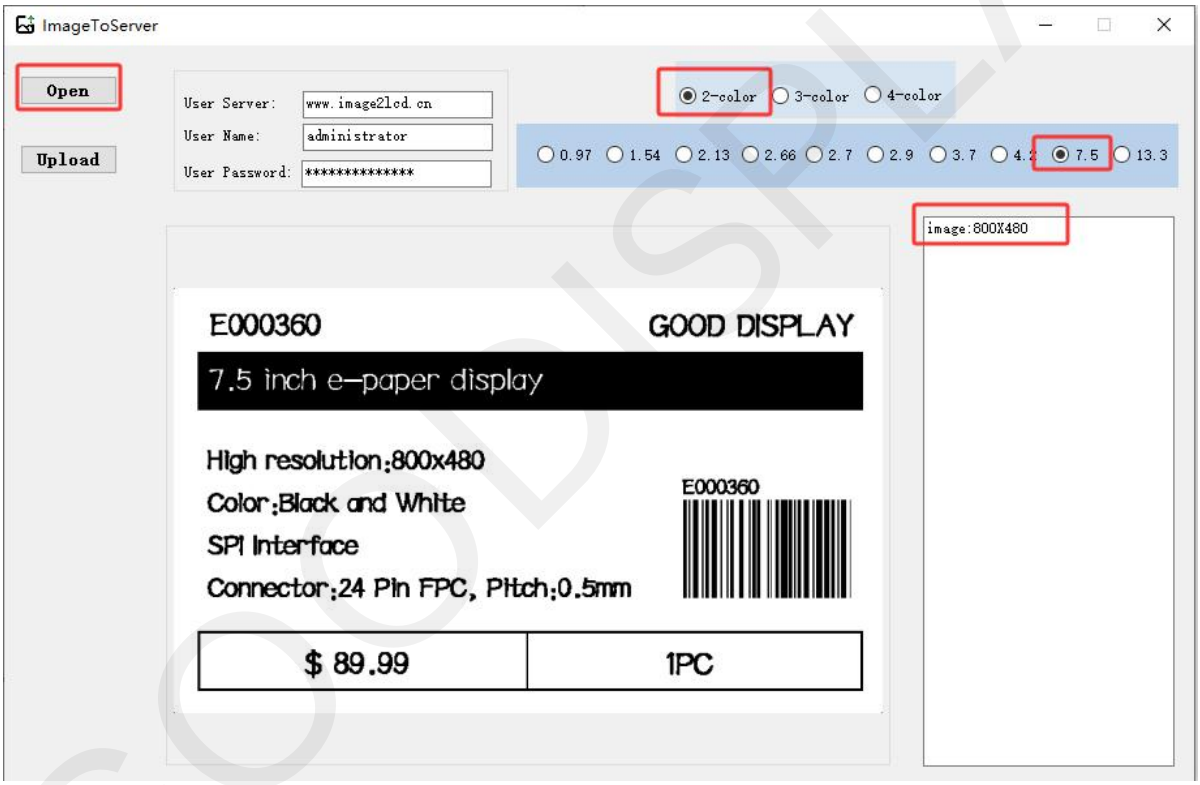
### 7.3 WiFi Network Settings

For the device WiFi network settings, please refer to Section 5.2 for USB network configuration on the PC side.

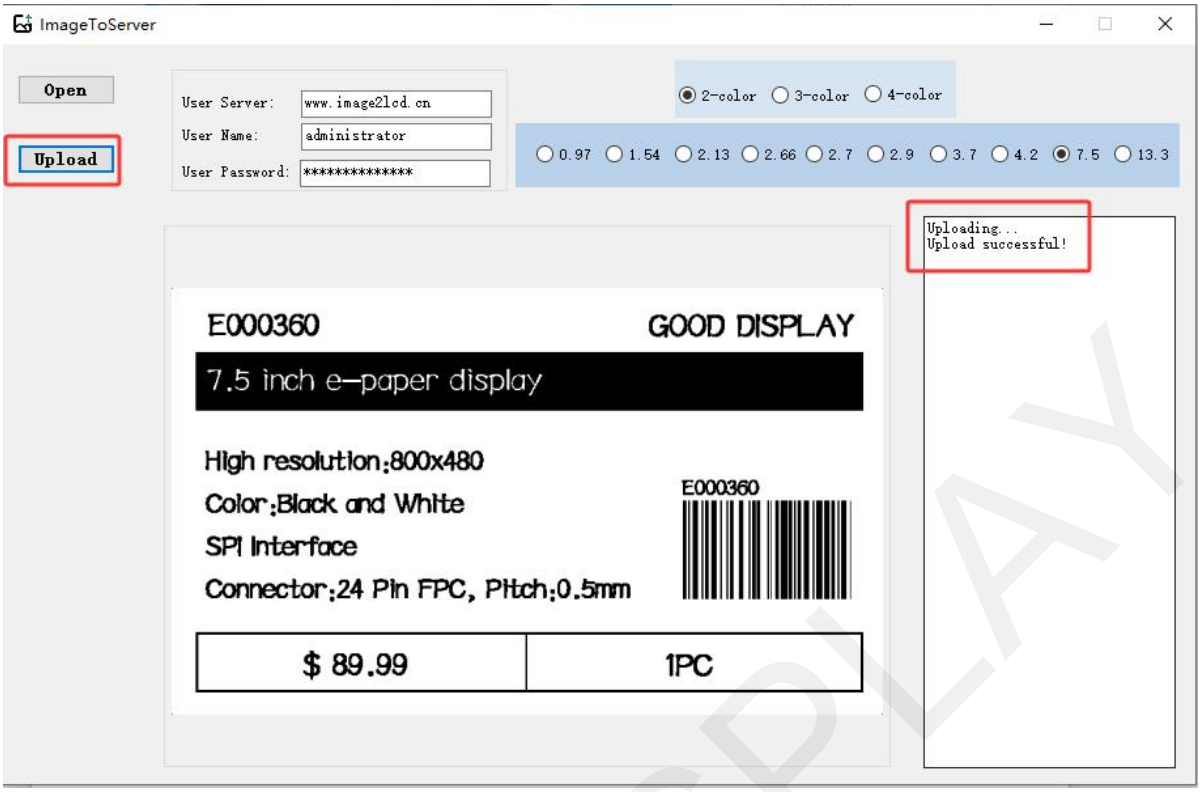
### 7.4 Image Display

1) Power on the driver board. It is recommended to use a power supply of 5V 1A or above.

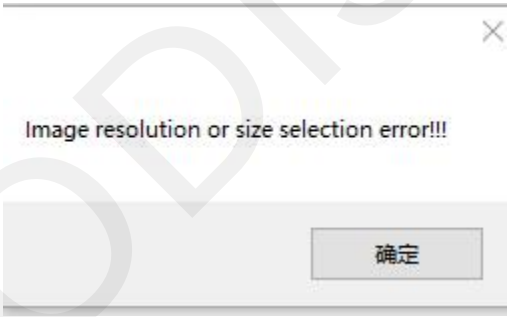
2) Open the ImageToServer software, click the "Open" button, and select the prepared image with a resolution of 800 × 480. After importing the image, the software will display the image size, resolution, and other parameters. Select "2-color" for color mode and "7.5" for screen size.




3) Click the button below  . If the image resolution matches the e-paper display size, the information panel on the right will display the current image upload status.




If a warning message pops up after clicking the upload button, you need to reselect the screen color or size and other related settings.




4) Wait for about 10 seconds. The device will automatically retrieve the image that was just uploaded and display it on the e-paper screen.